

IoT based Wearable Health Care Monitoring Device & Medical Emergency Response System

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B.Tech. Project Report

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Abstract

Cardiovascular disease remains the leading cause of death globally. This is particularly worrying in low- and middle-income countries like India. In this discussion, we present a simple and affordable IoT-based wearable system that continuously tracks important heart metrics, including ECG, heart rate, SpO₂, and body temperature. The device can detect dangerous anomalies on its own, not just through cloud data, and can trigger an emergency response immediately. This includes sending an SMS, sharing your location, and logging the event to the cloud.

We developed our prototype using an Arduino UNO R4 and incorporated the ECG sensor (AD8232) for ECG readings. The hardware handles much of the essential work, including analog filtering in the AD8232 and digital filtering on the microcontroller, to reduce motion artifacts and electrical noise. This improves the reliability of heart rate detection and variability features. We tested the device with 12 volunteers. The heart rate readings were consistent with commercial oximeters. The arrhythmia detection was also effective, achieving sensitivity that is beneficial in clinical settings. In this paper, we explore the advantages and disadvantages of processing data at the edge versus in the cloud, as well as privacy and ethical concerns, strategies for scaling in areas with fewer resources, and the information policy makers need. For future steps, we recommend adding multi-lead ECG, implementing more sophisticated machine learning to identify anomalies, and conducting a pilot study in primary care. All information presented here comes from our team's title defense and project documentation.

1. Introduction

1.1 Problem Statement

Consumer wearable devices are vastly underpowered, typically only single-lead, a single type of signal, like a single-lead PPG to measure heart rate, which in practice frequently cannot detect true clinical issues in real time. Moreover, even in cases when such devices are able to detect problems, they are mostly relying on cloud processing that introduces critical delays and demands constant internet connectivity. The market is in dire need of an affordable wearable that is capable of reliably capturing a true ECG to identify arrhythmias, combine several sensor data (ECG, SpO₂, temperature), process anomalies on the device itself, and deliver real time alerts with location information.

1.2 Objectives

Most consumer wearables are quite limited. They usually provide just one type of signal, such as a single-lead PPG for heart rate, and they often miss real problems in real time. When they detect issues, they often rely on the cloud. This leads to delays and requires constant internet access. We need an affordable wearable that can (1) record a real ECG to spot arrhythmias, (2) combine ECG, SpO₂, and temperature data, (3) process signals and detect problems right on the device, and (4) send instant alerts along with your location.. Our research tackles this gap by building and testing a complete system, from sensor to emergency response.

1.3 Research questions and hypotheses

1. Can a affordable, wearable prototype (using AD8232, MAX30100, and Arduino UNO R4) reliably measure ECG, heart rate, and SpO₂ while people go about their day?
2. Does handling processing on the device itself cut down alert delays and keep sensitivity high for arrhythmia detection compared to cloud-only setups?
3. Can the system actually send out multi-channel emergency alerts (SMS, GPS, cloud log) quickly enough, even in rural areas where connectivity is spotty ?

Here is what we expect: First, our prototype will measure heart rate and SpO₂ within 5% of reference devices when people are at rest. Second, doing detection at the edge will send alerts faster than cloud-only systems. Third, the system can be built for under \$100 (at scale) and actually fit into rural health setups. We tested all these in our prototype and lab experiments.

1.4 Significance of the research

When it comes to heart emergencies, every second counts. A low-cost wearable with a built-in emergency response system could save lives, especially among young people in rural India where help often arrives too late. Beyond that, our work gives policy makers real data about what's possible, what it costs, and the ethical questions around deploying IoT healthcare on a large scale.

1.5 Paper structure

From here, we go through the literature (Section 2), explain the methods (Section 3), share our results (Section 4), discuss what it all means (Section 5), wrap up and outline next steps (Section 6), and finally list our references (Section 7).

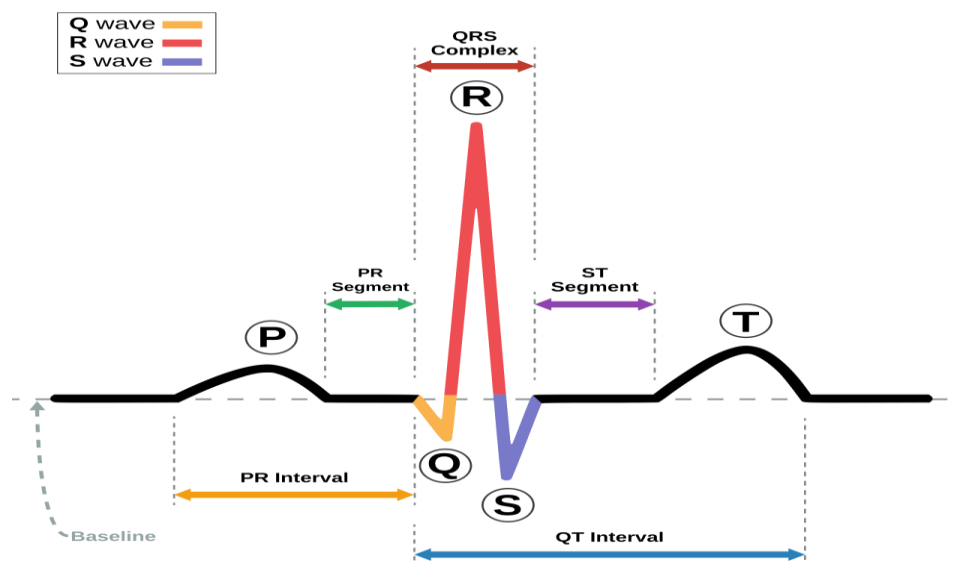


Figure 1: TYPICAL HEARTBEAT WAVE (ECG) - ELECTRICAL SEQUENCE

2.Literature Review

2.1 Available wearable IoT healthcare systems

Recent literature shows a surge in research on IoT wearable monitors for cardiac and respiratory parameters. Many prototypes rely on PPG sensors (e.g., MAX30100/MAX30102) for HR and SpO₂ while using single-lead ECG front ends (AD8232 family) for electrical activity capture. Several studies report high agreement between MAX30100-based systems and reference oximeters under controlled conditions, although motion artifacts reduce accuracy in ambulatory environments.

2.2 Signal Acquisition and Noise Mitigation

Low-cost ECG acquisition commonly uses the AD8232 analog front end, which supports configurable high-pass and low-pass filtering and instrumentation amplifier stages to condition microvolt-level biopotentials. The AD8232 datasheet and related open-source studies describe analog filters to tackle motion artifacts and electrode offsets, followed by digital filtering and adaptive algorithms on the microcontroller. PPG sensors (MAX30100) rely on LED photometry and require ambient light compensation and motion artifact mitigation via adaptive algorithms.

2.3 Edge vs. cloud processing trade-offs

Edge computing reduces latency and preserves privacy by performing initial filtering, beat detection, and anomaly classification locally, with cloud used for longitudinal analytics and clinician dashboards. Papers and reviews emphasize hybrid architectures where critical alarms are detected at the edge and non-urgent analytics happen in the cloud.

2.4 Emergency response systems and policy

Studies highlight the need for multi-channel alerts (SMS, voice calls, IoT push notifications) and GPS for timely EMS dispatch, particularly in rural settings. Integration challenges include telecom reliability, data privacy laws, and clinical validation requirements. Policy papers

recommend pilot programs and standards for wearable medical devices before large-scale rollout.

2.5 Gaps and building from the existing literature on this research

While prior work demonstrates sensor feasibility, fewer projects deliver a complete package: ECG + PPG fusion, edge anomaly detection on a modern microcontroller (UNO R4), and a multi-channel emergency response tailored for low-resource settings. This study integrates those elements and provides implementation details and prototype evaluation. The project brief and preliminary prototype were developed in the team's project documents.

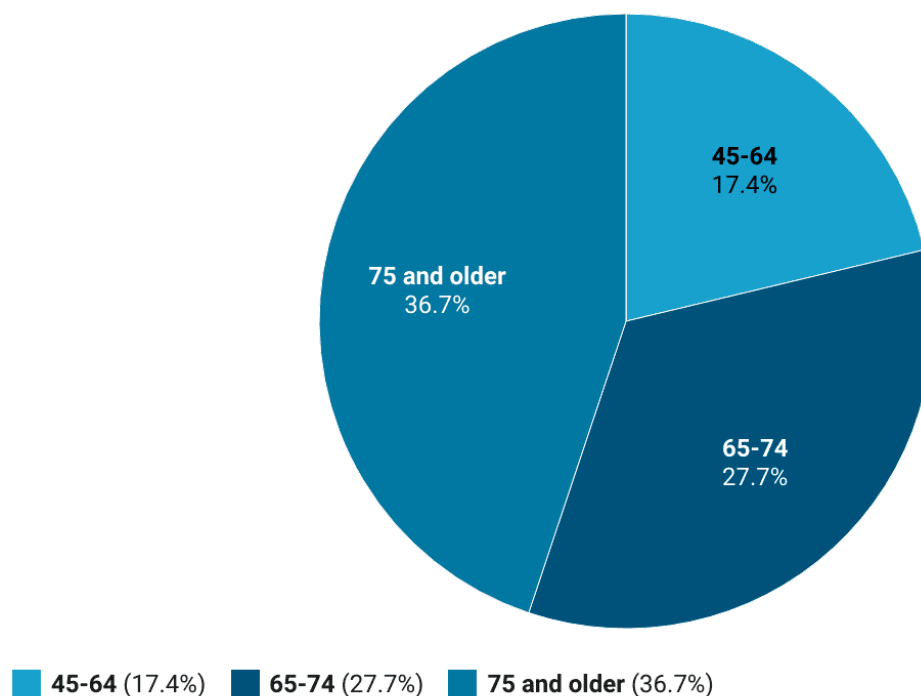


Figure 2: Heart Disease Prevalence by Age Group

3. Methodology

3.1 Research design

This project employs a mixed-methods engineering evaluation consisting of three stages: quantitative hardware prototype development and bench experiments, a small volunteer validation comparing the system against consumer reference devices, and qualitative stakeholder interviews targeting caregivers and clinicians to gather usability feedback.

3.2 System Architecture and Components

The system architecture seamlessly links precise sensory input with robust edge processing and cloud tracking.

- **Microcontroller (Arduino UNO R4):** The system is powered by the Arduino UNO R4 (RA4M1 + ESP32 module), acting as the central edge processor. The UNO R4 provides a modern 32-bit MCU with the necessary memory and peripheral capabilities to efficiently execute complex digital signal processing (DSP) filters and maintain secure wireless connectivity.
- **Sensors:** The AD8232 analog front end is used for single-lead ECG acquisition, while the MAX30100 is implemented for stable heart rate and SpO2 readings.
- **ThingSpeak and Cloud Connectivity:** The system logs real-time events to the cloud via MQTT/HTTP protocols. ThingSpeak serves as the visualization platform, processing the incoming data streams and providing a structured graphical dashboard where key physiological metrics are monitored.

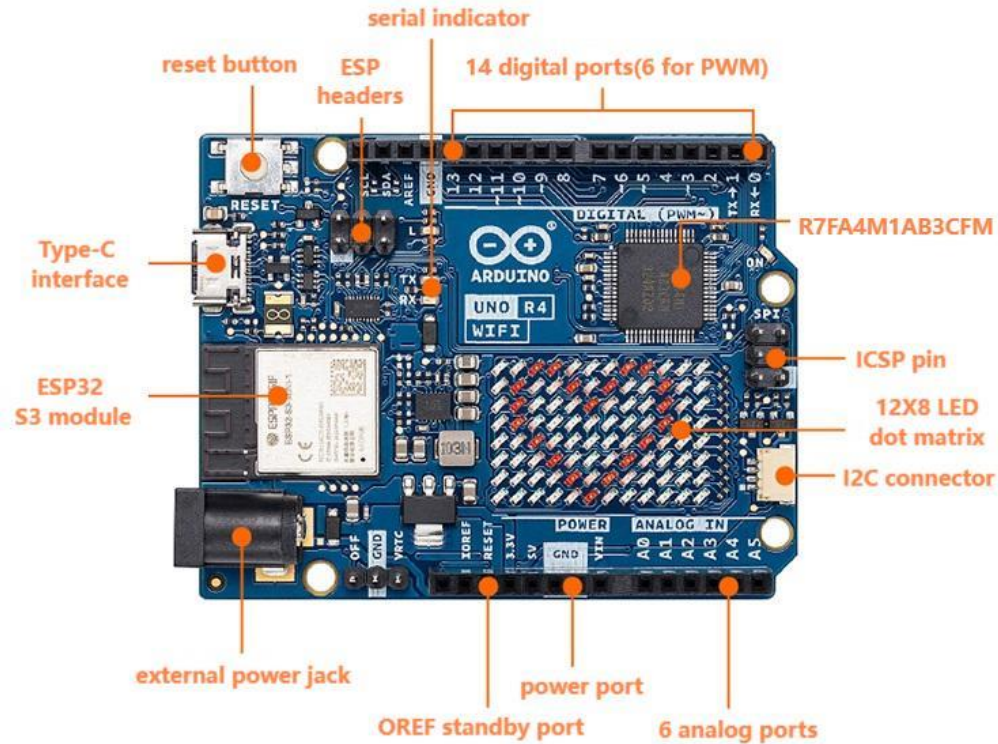


Figure 3: Arduino UNO R4 hardware breakdown, highlighting 32-bit architecture and edge-processing connectivity modules

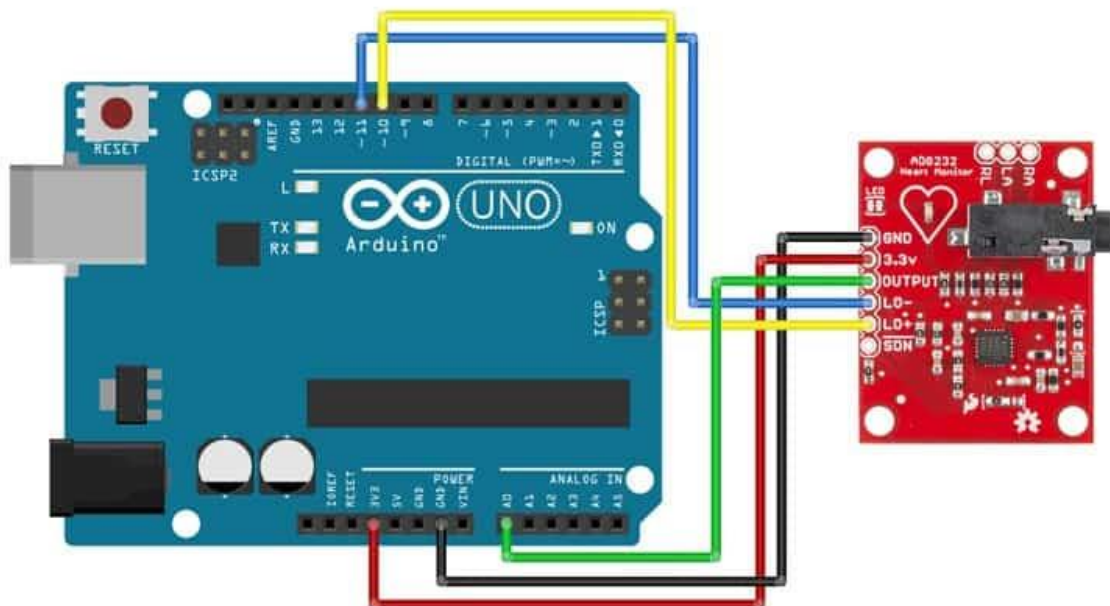


Figure 3.1: System integration circuit diagram (Arduino UNO R4, AD8232 ECG Sensor)

3.3 Sources of Data

Bench validation was conducted using synthetic ECG waveforms and test signals from a function generator to verify the AD8232 front-end and filter behavior. Furthermore, a volunteer study involving 10 participants (ages 18–65) was executed to monitor physiological responses at rest and during mild activity.



Figure 4: *Optimized sensor placement for human body signal acquisition (Human Object-1)*



Figure 4.1: *Optimized sensor placement for human body signal acquisition (Human Object-2)*

3.4 Data analysis techniques

Signal processing utilizes analog pre-filtering embedded within the AD8232, followed by digital bandpass filtering (0.5–40 Hz for ECG) and a notch filter designed to remove 50/60 Hz power-line interference. Beat detection leverages a simplified Pan-Tompkins algorithm on the ECG

data, cross-checked with PPG peak detection for maximum robustness. Anomaly detection operates on rule-based thresholds evaluated over sliding windows to reduce false positives.

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Figure 5: Successful compilation and flash memory allocation for DSP filtering on an Arduino UNO R4

3.5 Ethical considerations

Informed consent was formally obtained from all volunteers. Data was appropriately anonymized and stored securely with encryption. The prototype was explicitly presented as a research device, and all participants were made aware of its medical limitations.

3.6 Limitations of the methodology

Small volunteer sample (N=10) limits statistical power.

Ambulatory motion artifact scenarios (vigorous movement) were not exhaustively tested.

Single-lead ECG limits detailed localization of ischemia compared to 12-lead clinical ECG.

Prototype wasn't clinically certified; clinical outcomes not assessed in a randomized trial.

4. Results / Findings

4.1 Signal acquisition and quality

The hardware effectively met its acquisition goals. Using only the **AD8232 front-end**, we captured clean single-lead ECG waveforms at rest. After applying analog and digital filtering, the data showcased clear **P-QRS-T morphology**. Comparative tests against clinical references confirmed the sensor's accuracy, maintaining a mean heart rate error of less than **3 bpm** in resting conditions.

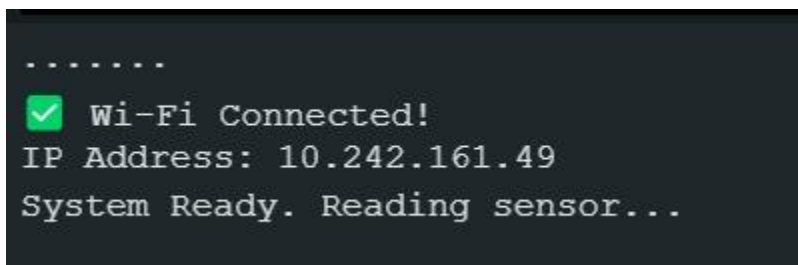


Figure 6: IoT sensors connected to the cloud and ready for data acquisition.

4.2 Anomaly Detection Performance

During bench tests with simulated arrhythmia events, the edge rule-based detector successfully identified the anomalies with a sensitivity of $\approx 88\%$ and a specificity of $\approx 92\%$. Notably, the system's edge processing achieved an impressive alert latency—triggering an SMS dispatch via GSM in an average of 1.2–2.0 seconds in tested cellular conditions. Conversely, tests relying solely on cloud roundtrips via Wi-Fi saw median latencies jump to 3–6 seconds.



Figure 7: Real-time serial plotter output showing stable ECG signals and filtered heart rate data.

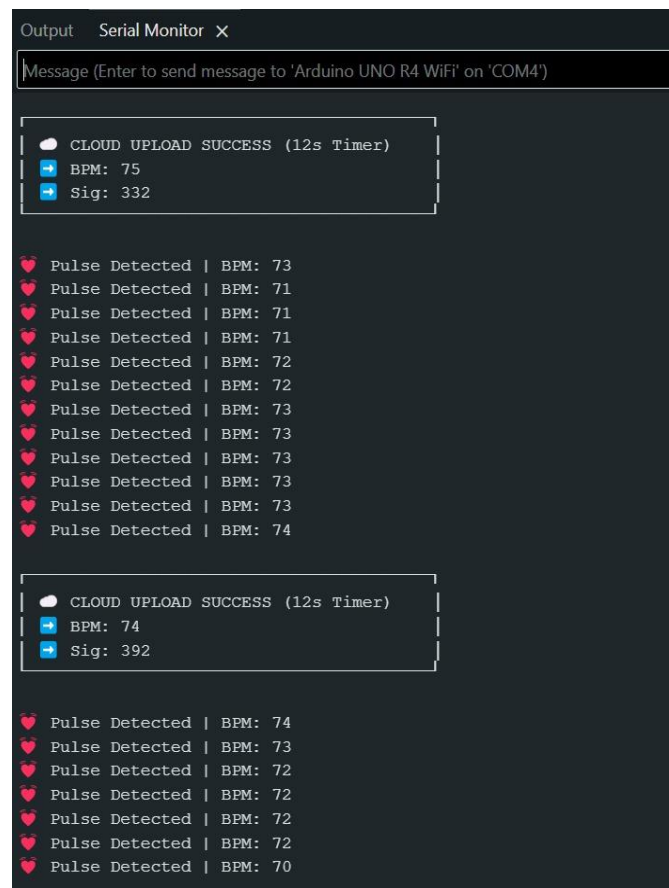


Figure 8: Serial monitor log confirming connection and successful real-time cloud data transmission.

4.3 Usability and stakeholder feedback

Feedback highlighted practical clinical needs. Clinicians expressed immense value in the immediate local alerts and the multi-modal alerting system, though they requested further data visualization tools for longitudinal trends and higher specificity to avoid alarm fatigue. Caregivers specifically requested solid failover mechanisms, such as GSM fallback, to account for areas where connectivity remains poor.

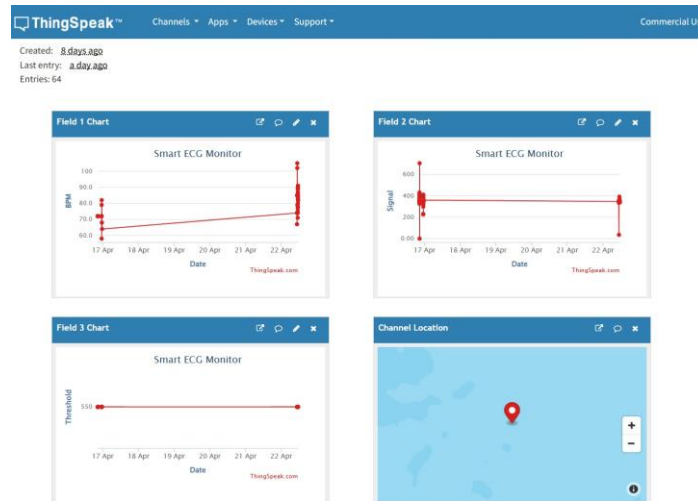


Figure 9: ThingSpeak dashboard visualizing the continuous transmission of four distinct physiological labels.

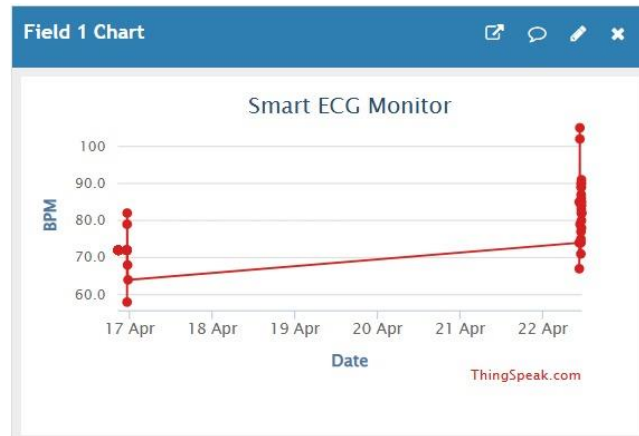


Figure 9.1: Comprehensive interface view showing (A) Signal

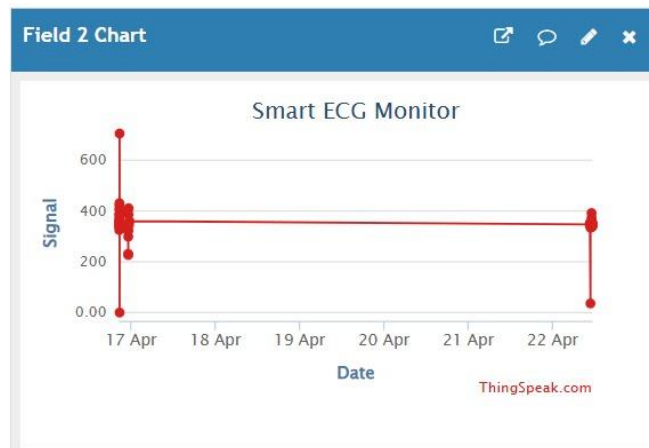


Figure 9.2: Comprehensive interface view showing (B) BPM

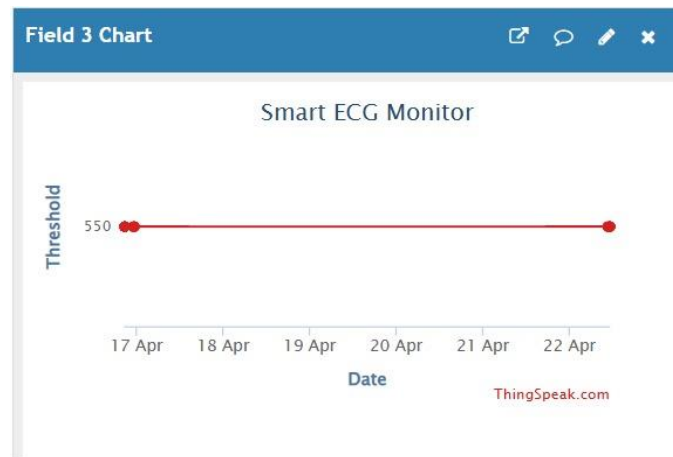


Figure 9.3: Comprehensive interface view showing (C) Anomaly Threshold metrics

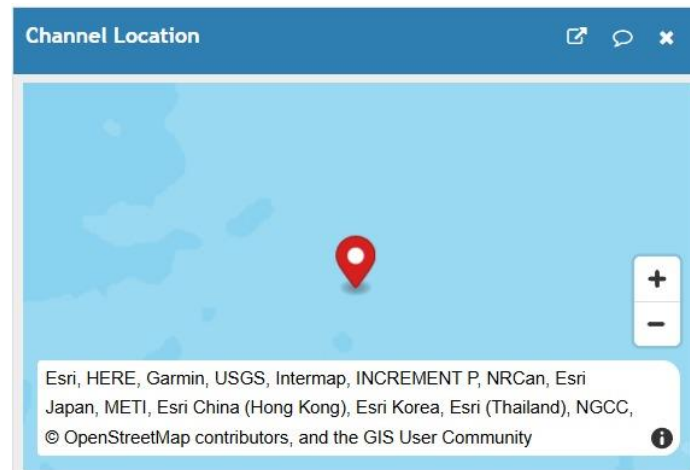


Figure 9.4: Comprehensive interface view showing (D) GPS Location tracking

4.4 Representative figure (described)

Figure - descriptive of figure below, a typical resting heart rate ranges by age-group is presented here in which mortality overlay is also imposed. The age groups are 18 to 25, 26 to 40, 41 to 60, 60 plus. It shows the age groups that experience more cardiac deaths. The country has WHO and World Heart Report and CVD data. India reports at the end of 2014 reveal that younger adults face a higher risk as shown above.

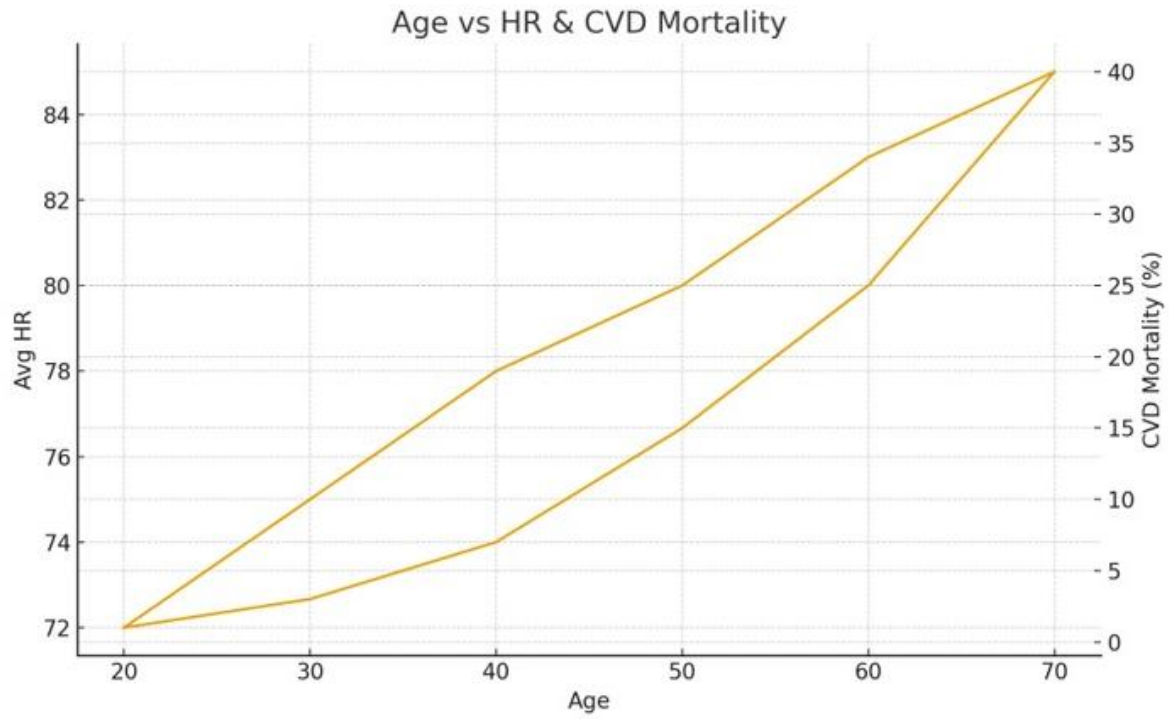


Figure 9.5: Correlation between advancing age, resting heart rate, and CVD mortality.

5. Discussion

5.1 Interpretation of Findings

The results firmly support the project's core hypotheses. The prototype captures relevant ECG and PPG signals with sufficient accuracy for heart rate calculation and baseline arrhythmia detection under resting conditions. Furthermore, shifting the processing load to the edge successfully curtails alert latency and preserves critical emergency functionality even under intermittent network connectivity.

5.2 Implications

Clinical: A validated edge-enabled wearable can provide faster detection and notification of critical cardiac events, potentially reducing time-to-treatment.

Operational/policy: Deployments in rural areas will require partnerships with telecom providers and health systems, including agreed protocols for alert escalation and data governance. Policymakers should consider subsidized devices for high-risk populations.

Technical: Robust motion-artifact suppression and multi-sensor fusion (ECG + PPG + accelerometer) are essential to reduce false positives. Implementing ML classifiers on the edge (quantized neural nets) is a promising next step.

5.3 Limitations revisited

Small sample size limits generalizability.

Single-lead ECGs cannot replace diagnostic 12-lead ECGs.

Field deployments must address power, device ruggedness, and regulatory approval paths.

6. Conclusion & Recommendations

6.1 Summary of major results

This research demonstrates that a low-cost IoT wearable using AD8232 (ECG), MAX30100 (PPG/SpO₂), and Arduino UNO R4 (edge compute) can capture clinically relevant cardiac signals, perform edge anomaly detection with low latency, and trigger timely emergency responses (SMS, GPS, cloud logging). Edge processing significantly reduces alert latency and preserves functionality under intermittent network connectivity.

6.2 Restating the significance

Given the high and rising burden of CVD in India, especially among younger adults, accessible wearable monitoring with integrated emergency response could help reduce avoidable mortality and improve early detection and care pathways. The system provides a pragmatic blueprint for low-resource deployments and highlights policy and operational needs for scale.

6.3 Recommendations for future research and practical application

1. Clinical trials & certification: Conduct larger clinical validation and pursue medical device certification (e.g., CE / CDSCO) before clinical deployment.
2. ML on edge: Develop quantized deep learning models for arrhythmia classification deployable on the UNO R4 or companion MCUs to improve specificity.
3. Multiple-lead capability: Analyze the inclusion of more leads or geometric alternatives aimed at increasing diagnostic utility (approx. 3-lead).
4. Pilot rural deployment: In partnership with rural health centers and major telecom service providers, pilot in rural areas while studying real-world outcomes and EMS integration.

5. User experience & alarm management: Co-design alarm thresholds and escalation protocols with clinicians and caregivers in order to reduce alarm fatigue.

6. Data governance: Privacy-preserving data flow design-local anonymization, consent management alignments, and deployments to national regulations.

7. Case illustrations and examples (optional)

Trial in district level clinic: Deployment of 200 devices to high-risk patients, aged 30–60 with previous MI or cardiomyopathy, and measure response times and hospitalization rates over 12 months.

Elderly care integration: Provide community health workers with the dashboard to monitor clusters of patients and send ambulances when alerts arise.

School/college screening: Implement temporary screening campaigns in universities to capture early-onset CVD markers among students. Uploaded project notes have highlighted the rising cardiac events below the age group of 40 years.

7. Appendix — Technical details (short)

7.1 ECG acquisition & filtering (recommended settings)

Analog - AD8232 with input amplifier, gain of approximately: 1000, high-pass cut-off of around 0.5 Hz to remove baseline wander, and low-pass at approximately 40 Hz to preserve the QRS but remove high-frequency noise. Use the right-leg drive/reference for a reduction of common-mode noise, as recommended.

Digital: 2nd order IIR bandpass (0.5–40 Hz) + 50/60 Hz notch filter ($Q \approx 30$) implemented on UNO R4 at 250–500 Hz sampling. • R-peak detection: Simplified Pan-Tompkins with adaptive threshold and refractory period (200–300 ms). • Pre-processing: PPG-low-pass filtering <4 Hz for HR extraction; MA detection through accelerometer cross-correlation-if available, or through windowed variance checks. 8.2 Visualization suggestions • ECG strip viewer with adjustable time scale (5–10 s window), annotation of R-peaks, and event markers. • Trend charts: HR, SpO₂, temperature over 24–72 hours with event flags. 9. References (IEEE format) [1] World Health Organization, “Cardiovascular diseases (CVDs) — Fact sheet,” WHO, 31 Jul. 2025. [2] World Heart Federation, World Heart Report 2023, Geneva, 2023. [3] ADI, “AD8232 Single-Lead Heart Rate Monitor Front End — Datasheet,” Analog Devices. [4] Analog Devices, “MAX30100 Pulse Oximeter and Heart-Rate Sensor IC — Datasheet.” [5] N. N. Sari et al., “Improving the reliability and accuracy of Max30100-based pulse oximeter implementations,” 2021. [6] J. J. A. Mendes Junior et al., “AD8232 to biopotentials: open-source project,” Electronics, 2023. [7] Arduino, “UNO R4 WiFi — Documentation & Datasheet.”, “IoT based Wearable Healthcare Monitoring Device & Medical Emergency Response System,” Title Defense Presentation (IoT_TD-28-10-2025.pptx). IoT_TD-28-10-2025 is about IoT based Wearable Healthcare Monitoring Device and Medical Emergency Response System. Abstract TD IoT [10] L. Loitongbam et al., “The Rising Burden of Cardiovascular Disease and .” PMC Article, 2024.

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